llmXive Automated Review of Redesign Mixture-of-Experts Routers with Manifold Power Iteration

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript makes several strong claims regarding theoretical guarantees and empirical generalization that are not fully supported by the provided evidence or derivations.

First, the Introduction explicitly lists “providing a theoretical analysis yielding tighter bounds on expert utilization” as a main contribution. However, the “Methodology” and “Method Analysis” sections focus on deriving an optimization interpretation (Eq. 10) and demonstrating empirical alignment (Table 1). There are no formal mathematical bounds presented that quantify expert utilization (e.g., variance in expert load or collision probability) relative to standard routers. The claim of “tighter bounds” is therefore unsupported by the text.

Second, the theoretical derivation in Section 3.2 (Eq. 10) and Appendix A asserts that the proposed update rule is an “approximation” of steepest ascent on the manifold. While the structural similarity is noted, the paper claims a “striking structural alignment” and implies equivalence without deriving the error term of the approximation or proving that the approximation error vanishes under the non-stationary dynamics of online training. The claim that the method is “equivalent to a steepest ascent optimization” (Introduction) is too strong given the lack of a rigorous error analysis.

Third, the claim in Section 4.1 that the design is “agnostic to shifts in model features across optimizers” is based solely on 1B parameter experiments. While the results are promising at this scale, asserting fundamental agnosticism for larger scales (11B+) without theoretical justification or larger-scale ablation studies constitutes an overgeneralization of the empirical evidence.

Finally, the efficiency claim in Section 4.2 states the overhead is “negligible” and “does not exceed that of N extra tokens.” The power iteration step involves computing \$R \times W \times W^T\$, which is a matrix-matrix operation significantly more expensive than the vector-matrix operations in standard routing. The text lacks a specific arithmetic breakdown or complexity analysis to substantiate the claim that this cost is equivalent to adding \$N\$ tokens, making the “negligible” assertion difficult to verify.

Action items.

- **[writing]** In Section 1 (Introduction), the claim that the method ‘provides a theoretical analysis yielding tighter bounds on expert utilization’ is unsupported. The paper derives an optimization interpretation (Eq. 10) and convergence intuition but presents no formal bounds on utilization metrics (e.g., load variance or collision rates) compared to baselines. This overstates the theoretical contribution.
- **[writing]** In Section 3.2 (Eq. 10) and Appendix A, the derivation claims the update rule is an ‘approximation’ of steepest ascent. However, the text asserts ‘striking structural alignment’ and ‘equivalence’ without quantifying the error term or proving the approximation holds under the specific non-stationary conditions of online training. The claim of equivalence is too strong given the lack of error bounds.
- **[writing]** In Section 4.1, the claim that the method is ‘agnostic to shifts in model features across optimizers’ is based on 1B parameter experiments. Extrapolating this to claim fundamental agnosticism for 11B+ models without explicit theoretical justification or larger-scale ablation is an overgeneralization of the evidence provided.

- **[writing]** In Section 4.2, the claim that the 0.2% slowdown is 'negligible' and 'does not exceed that of N extra tokens' lacks a rigorous breakdown. The power iteration involves matrix-matrix multiplications ($R imes W imes W^T$) which are computationally heavier than simple token additions. The specific arithmetic justification for this claim is missing from the text.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure displays clear convergence curves with error bands, but the caption contains formatting artifacts and the legend fails to explicitly link the plotted lines to the specific 'router design' claim.

Figure 2

The figure displays loss and accuracy curves but fails to support the 'faster convergence' claim as the x-axis represents tokens rather than time. Additionally, the caption contains a text placeholder error.

Figure 3

The figure is visually clear but suffers from a critical caption error where the method name is missing, leaving the purple line undefined. Additionally, the y-axis scale is zoomed in significantly, potentially exaggerating the visual impact of the load balancing loss reduction.

Figure 4

The figure is fundamentally misleading as the data shown (high accuracy for the 'w.o Retraction' condition) directly contradicts the caption's claim of 'pretraining collapses'. Additionally, the token scale on the x-axis appears inconsistent with the 3B model size mentioned in the caption.

Figure 5

The figure effectively visualizes loss convergence across three optimizers, but the caption contains a broken placeholder ('\\') and the top subplot's legend ('AdamW') contradicts the caption's list of optimizers ('AdamW, AdamH, Muon'), creating confusion about which optimizer is being compared in the top panel.

Action items.

- [**writing**] Figure 1: The caption contains a placeholder artifact 'with ,' and the model name 'grayMuonH-1B' appears to be a raw filename or variable name rather than a clean description.
- [**science**] Figure 1: The legend labels 'MuonH' and 'MuonH w. MPI' do not explicitly identify which line corresponds to the 'router design' mentioned in the caption, making the claimed 0.013 reduction difficult to verify visually.
- [**science**] Figure 2: The caption claims the method facilitates 'faster convergence,' but the x-axis is 'Tokens' (compute budget), not 'Wall-clock time' or 'Steps.' Since the curves overlap significantly in the loss plot (a), the claim of faster convergence is not supported by the provided axes.
- [**writing**] Figure 2: The caption contains a missing variable placeholder (a backslash '\') before 'facilitates,' likely referring to the specific router design or method name which is absent from the text.
- [**fatal**] Figure 3: The caption states 'Load balancing loss for 3B MoE with [missing method]', but the rendered plot shows two distinct lines ('MoE' and 'MoE w. MPI') without defining what the purple line represents in the text. The caption is incomplete and fails to describe the comparison shown.
- [**science**] Figure 3: The y-axis scale is extremely narrow (0.016 to 0.017), which visually exaggerates the difference between the two curves. While the absolute difference is small, the visual presentation may mislead readers regarding the magnitude of the improvement in load balancing loss.
- [**fatal**] Figure 4: The caption claims to show 'pretraining collapses' without Router Retraction, but the plot displays 'Accuracy (%)' on the y-axis, not pretraining loss. Furthermore, the 'MPI w.o Retraction' line (pink) shows high accuracy (~53%) and does not collapse, directly contradicting the caption's claim of instability.
- [**science**] Figure 4: The x-axis labels (100B, 125B, 150B, 175B, 200B) imply a 200B token training run, but the caption specifies this is for a '3B MoE' model. This scale is inconsistent with the model size described in the caption.
- [**writing**] Figure 4: The caption mentions 'Router Retraction' and 'Power Iteration' as key design choices, but the legend labels ('MoE w. MPI', 'MPI w.o Power-Iter', 'MPI w.o Retraction') are inconsistent in naming convention and do not clearly map to the specific ablation conditions described.
- [**writing**] Figure 5: The caption contains a placeholder '\\' instead of the specific method name (e.g., 'MuonH' or 'MPI') that the figure legends refer to as 'w. MPI', making the claim 'MoE with \\ achieves...' unreadable.

- **[writing]** Figure 5: The top subplot legend lists ‘AdamW’ and ‘AdamW w. MPI’, but the caption claims a comparison across ‘AdamW, AdamH, Muon’; the top plot should likely be labeled ‘AdamH’ to match the caption’s description of the three optimizers.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and jargon that are either undefined, conflicting, or prone to confusion for a broader audience. The most critical issue is the definition of **MPI**. In the preamble (line 42), `MPI` is explicitly defined as “Message Passing Interface,” a ubiquitous standard in parallel computing. However, the authors use `MPI` throughout the text to refer to their proposed “Manifold Power Iteration.” This contradiction will inevitably confuse readers, particularly those from systems backgrounds, and must be resolved by removing the conflicting preamble definition and ensuring the paper’s specific acronym is defined clearly at first use in the abstract or introduction.

Furthermore, several terms are used without sufficient context for non-specialists. The term **“midtrain”** (Section 5.2) is industry slang for continued pretraining; replacing this with “continued pretraining” would improve clarity. The metric **“MaxVio”** (Section 5.2, Table 4) is introduced without definition; the full name (e.g., Maximum Violation) and a brief explanation of what it measures are required. Additionally, **“Hyperball Optimization”** is mentioned with a typo (“Opimization”) and without a plain-language explanation of the geometric constraint it imposes. Finally, the phrase “Principle Singular direction” (Section 6.1) should be corrected to “Principal” to align with standard linear algebra terminology. Addressing these points is essential to ensure the paper is accessible to readers outside the immediate sub-field of MoE router optimization.

Action items.

- **[writing]** The acronym ‘MPI’ is defined in the preamble as ‘Message Passing Interface’ (line 42), conflicting with the paper’s usage for ‘Manifold Power Iteration’. Remove the conflicting preamble definition and define ‘Manifold Power Iteration’ clearly at first use in the abstract or introduction.
- **[writing]** Replace the industry slang ‘midtrain’ (Section 5.2) with ‘continued pretraining’ to ensure clarity for non-specialist readers unfamiliar with this specific training phase terminology.
- **[writing]** Define the metric ‘MaxVio’ (Section 5.2, Table 4) at first use. The text mentions it reflects load balance but does not define the acronym (e.g., Maximum Violation) or its calculation.
- **[writing]** Correct the typo ‘Opimization’ to ‘Optimization’ in Section 5.1 and briefly explain

‘Hyperball Optimization’ (e.g., constraining weights to a hypersphere) for readers unfamiliar with the cited work.

- [**writing**] Correct the grammatical error ‘Principle Singular direction’ to ‘Principal Singular direction’ in Section 6.1 to align with standard linear algebra terminology and avoid confusion.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The logical consistency of the paper is generally sound in its high-level narrative: the motivation (router-expert misalignment) leads to a proposed solution (aligning with principal singular directions via power iteration), which is then empirically validated. However, specific causal claims and theoretical equivalences lack rigorous support, creating gaps between premises and conclusions.

First, in Section 3.3 (“From Maximum Projection Constraints to Manifold Power-Iteration”), the authors claim their update rule is “equivalent to a steepest ascent optimization.” The derivation in Equation 10 shows that the update direction $\Delta \vr_M$ shares a structural form with the projected gradient $\Delta \vr_g$. However, the paper treats this structural similarity as mathematical equivalence. The adaptive step-size in $\Delta \vr_M$ (the denominator term) is derived from the power iteration process, not from an optimization line-search or manifold geometry argument. Asserting equivalence without proving that this specific step-size yields the same trajectory or convergence rate as true steepest ascent is a logical leap. The conclusion that the method is “tailored for maximum projection constraints” is plausible, but the claim of equivalence is unsupported.

Second, the explanation for the failure of 10 power iterations (Section 4.1) is logically weak. The authors state that “aggressive alignment disrupt the stability of router optimization.” Standard power iteration theory suggests that more iterations lead to a more accurate estimate of the principal singular vector, which should theoretically improve the alignment objective. The observed performance drop (loss increase, downstream drop) is attributed to “stability,” but no mechanism is provided. Why would a more accurate alignment with the expert’s dominant direction destabilize the training? Is it due to overfitting the current batch’s expert weights? Is it a gradient variance issue? Without a theoretical or empirical analysis of the gradient dynamics or spectral properties under multiple iterations, the causal link between “tighter alignment” and “instability” remains an assertion rather than a derived conclusion.

Third, the attribution of load balancing improvements to the “retraction design” (Section 4.1) is contradicted by the ablation results. The authors claim the improved balance is an “unexpected bonus” of the retraction. However, Figure 4 and the accompanying text note that the “Retraction-only” baseline (which performs normalization without the power iteration step) exhibits a “similar balance loss distribution” to the full MPI method. If the retraction alone achieves the balance, the causal claim that the *combination* of power iteration and retraction (the core novelty) is responsible for the load balancing benefit is weakened. The data suggests the normalization constraint is the primary driver, yet the paper frames the entire "Power-then-Retract" paradigm as the cause.

These issues do not invalidate the empirical results but require clarification to ensure the theoretical claims and causal attributions are logically supported by the presented evidence.

Action items.

- [**science**] The claim that MPI is 'equivalent to a steepest ascent optimization' (Sec 3.3) is unsupported. Eq. 10 shows structural similarity, not equivalence. The adaptive step-size is not proven to match the optimal step-size for steepest ascent on the manifold, weakening the theoretical claim.
- [**science**] The argument that 'aggressive alignment disrupts stability' (Sec 4.1) with 10 iterations contradicts standard power iteration convergence. The paper lacks a mechanism (e.g., gradient variance analysis) to explain why tighter alignment to the principal direction would destabilize the optimizer.
- [**science**] The load balancing improvement is attributed to 'retraction design' (Sec 4.1), yet the ablation (Fig 4) shows the 'Retraction-only' baseline achieves similar balance. This suggests normalization, not the power iteration interaction, drives the benefit, weakening the causal claim for the full method.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The manuscript exhibits significant overreach in its theoretical and comparative claims relative to the provided evidence.

First, the Introduction explicitly lists as a main contribution: "(2) we provide a theoretical analysis yielding tighter bounds on expert utilization." However, a review of the Methodology and Appendix reveals no such bounds. Section 3.3 derives an update rule approximation and discusses convergence to the principal singular vector, but it never establishes a bound on *expert utilization* (e.g., load balancing metrics or utilization rates) nor proves it is "tighter" than existing methods. This is a clear over-claim of theoretical contribution.

Second, the Abstract and Introduction assert that the approach "matches or exceeds" recent alternatives such as the Switch Transformer. Yet, Section 5.2 ("Comparative Analysis with vanilla MoE") explicitly states: "We forgo comparisons with other baselines since our design conforms to the standard router form and is theoretically orthogonal to these studies." The paper only compares against a "vanilla MoE" baseline (presumably a standard router implementation) and does not include Switch Transformer or other named SOTA routers in the experimental tables. Claiming superiority over specific external baselines without presenting data for them is unsupported.

Finally, the claim of “zero inference overhead” in Section 5.2 is absolute and potentially misleading. While the authors argue weights can be pre-computed, the method requires computing $R_{[i]} W_g^i W_g^{i\top}$ (Eq. 2). If this is done at load time, it incurs a startup cost dependent on the expert matrix size, which is non-zero. If done online, it adds computation. The text should clarify the trade-off rather than asserting zero overhead.

These issues require revision to align the claims strictly with the presented data and derivations.

Action items.

- [**science**] The Introduction claims the method provides ‘tighter bounds on expert utilization’ (Section 1, Contributions), yet the paper presents no formal bounds, theorems, or proofs regarding utilization rates. This is an unsupported theoretical claim that must be removed or substantiated with actual derivations.
- [**science**] The Abstract and Introduction state that the method ‘matches or exceeds’ recent alternatives like Switch Transformer, but the Experiments section (Section 5) explicitly states: ‘We forgo comparisons with other baselines since our design conforms to the standard router form.’ The claim of superiority over specific named baselines is unsupported by the provided data.
- [**writing**] The paper claims ‘zero inference overhead’ (Section 5, Efficiency Analysis) because weights are pre-computed. However, the ‘Power-then-Retract’ step requires matrix-vector products with expert weights ($W_g W_g^T$) during the forward pass (Eq. 2) or at load time. If computed at load time, this adds a non-trivial initialization cost proportional to expert size, which contradicts the ‘zero overhead’ absolute claim. If computed online, it adds FLOPs. The claim needs qualification.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript addresses safety and ethics primarily through a dedicated ‘Ethical Considerations’ section (lines 13-24) and a ‘Broader Impact’ section (lines 1085-1093). The authors correctly identify the dual-use nature of their work, noting that improved efficiency in Mixture-of-Experts (MoE) models could lower barriers for generating disinformation or malicious content. They also appropriately mention data privacy regarding the use of public benchmarks.

However, the current treatment of these issues remains high-level and generic. The ‘Ethical Considerations’ section advises practitioners to “adopt usage policies” and “employ appropriate anonymisation techniques” but does not propose or evaluate specific technical mitigations that could be integrated into the router design itself or the deployment pipeline. Given the paper’s focus on a fundamental architectural change, a more concrete discussion on how this specific router design might interact with safety mechanisms (e.g., does the power iteration step affect the model's ability to be aligned or watermarked?) would be valuable.

Furthermore, the 'Reproducibility' section (lines 1065-1072) and the code availability statement in the preamble (line 1067) contain placeholder URLs (`https://github.com/yourorg/mixture-of-experts-routers`) and a non-functional commit hash. For a method claiming to improve training stability and efficiency in large-scale models, the inability to inspect the actual implementation prevents the community from auditing the code for potential safety-critical bugs, numerical instabilities, or unintended side effects that could arise during deployment. While external code hosting is acceptable, the links provided must be valid and point to the actual artifacts used in the experiments.

Finally, while the authors state that training data lacks PII, there is no discussion regarding the risk of the model memorizing sensitive information from the broader pretraining corpus (FineWeb-Edu) or the evaluation datasets (Math/Code). A brief acknowledgment of memorization risks and any steps taken to mitigate them would align better with current best practices in responsible AI research.

Action items.

- **[writing]** The 'Ethical Considerations' section (lines 13-24) and 'Broader Impact' section (lines 1085-1093) acknowledge dual-use risks but lack concrete mitigation strategies. Explicitly detail proposed technical safeguards (e.g., specific watermarking techniques, access control protocols) or policy frameworks the authors intend to support, rather than generic calls for 'usage policies.'
- **[writing]** The 'Reproducibility' section (lines 1065-1072) and code availability statement (line 1067) cite a placeholder GitHub URL ('https://github.com/yourorg/mixture-of-experts-routers') and a generic commit hash. For a paper claiming efficiency gains in large-scale training, the absence of a verifiable, functional code repository prevents independent safety auditing of the implementation for potential instability or unintended behaviors.
- **[writing]** The manuscript states training data (FineWeb-Edu) contains no PII (lines 1078-1080). However, it does not address the potential for the model to memorize and regurgitate sensitive information present in the broader pretraining corpus or the specific 'Math' and 'Code' datasets used for evaluation. A brief discussion on memorization risks or privacy-preserving training techniques would strengthen the safety profile.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The manuscript presents a compelling theoretical motivation for aligning router weights with the principal singular directions of expert matrices. However, the scientific evidence supporting the empirical claims requires strengthening to meet the standards of rigorous machine learning research.

The primary concern is the absence of statistical significance testing. The paper reports performance improvements across multiple benchmarks (e.g., Table 1, Table 3) based on single training runs. In the context of large-scale pretraining (350B tokens), performance metrics can fluctuate due to random seed initialization, data shuffling, and optimizer dynamics. Without reporting standard deviations over multiple seeds (e.g., 3-5 runs) or providing confidence intervals, it is impossible to determine if the observed gains (e.g., +1.39% on GSM8K in Table 3) are statistically significant or merely artifacts of random variance. The claim of "faster convergence" in Figure 2 is similarly anecdotal without error bands.

Furthermore, the ablation studies, while present, are not fully conclusive regarding the specific contributions of the "Power" and "Retract" components. The authors demonstrate that removing retraction leads to instability (Section 5.2.1), which is a valid finding. However, to isolate the benefit of the power iteration itself, a comparison against a baseline that includes power iteration but lacks the specific retraction logic (if stable) or a more rigorous analysis of the "normalization-only" baseline is needed. The current evidence suggests the retraction is necessary for stability, but the marginal gain of the power iteration step over a simple normalized router (if the latter were stable) is less clearly quantified in the final results.

Finally, the efficiency claim of a "0.2% slowdown" is stated with high precision but lacks methodological detail. Was this measured as an average over the entire training run? What was the variance? Given the complexity of MoE training, such a small difference could easily fall within the noise of system-level fluctuations. Providing a brief description of the measurement protocol or error margins would bolster the credibility of this claim.

In summary, while the method appears promising, the current evidence relies on single-run point estimates that do not robustly support the strong claims of superiority and efficiency.

Action items.

- [**science**] The claim of 'faster convergence' and 'superior performance' lacks statistical significance testing. Table 1 (tab:opt-agnostic) and Table 3 (tab:midtrain) report single-point averages without standard deviations, confidence intervals, or p-values. Given the stochastic nature of LLM pretraining, single-run comparisons are insufficient to rule out random variance as the cause of observed gains.
- [**science**] The ablation study in Section 5.2.1 (lines 630-650) compares the full method against a 'normalization-only' baseline but fails to include a 'power-iteration-only' (no retraction) baseline in the final performance tables. While the text mentions instability for the latter, quantitative data on its performance (if stable runs were possible) or a clear explanation of why it cannot be compared fairly is needed to isolate the specific contribution of the retraction step versus the power iteration.
- [**science**] The efficiency claim of '0.2% slowdown' (Section 4.3, line 560) is presented as a definitive metric but lacks the necessary context of variance or measurement methodology. Was this measured over a single step or an average over the entire training run? Without

error bars or a description of the measurement protocol, this precise figure is not robustly supported.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The manuscript presents a novel router design for Mixture-of-Experts (MoE) models, but the statistical rigor of the empirical evaluation is insufficient to support the strong claims of superiority. The primary concern is the complete absence of measures of statistical uncertainty. Throughout the “Experiment” section (Sections 4 and 5), performance metrics such as downstream accuracy (Table 1, Table 3), perplexity (Table 2), and convergence rates (Figures 1-3) are reported as single point estimates derived from what appears to be a single training run per configuration.

In deep learning research, training dynamics are inherently stochastic due to random weight initialization, data shuffling, and optimizer noise. Reporting a single run without standard deviations, confidence intervals, or p-values makes it impossible to determine if the observed improvements (e.g., the 0.013 loss reduction in Figure 1 or the 2.33% accuracy gain in Table 3) are statistically significant or merely artifacts of a favorable random seed. For instance, the claim in Section 5.1 that increasing power iterations to 10 results in a “downstream drop of 1.39 percentage points” is presented as a definitive finding, yet without variance estimates, this difference could easily fall within the noise floor of the training process.

Furthermore, the sensitivity analysis of the hyperparameter $C’$ (Table 4) relies on single-point validation perplexity values. To robustly claim that the method is “relatively insensitive” to this parameter, the authors must demonstrate that the performance variance across seeds is small relative to the differences between $C’$ settings. The current presentation lacks the necessary statistical evidence to distinguish between genuine robustness and random fluctuation.

To address these issues, the authors should re-run their key experiments (main comparisons, ablation studies, and sensitivity analyses) with at least three independent random seeds. They must then report the mean and standard deviation (or 95% confidence intervals) for all quantitative results in tables and figures. Additionally, for the main performance claims, a statistical significance test (e.g., a paired t-test or Wilcoxon signed-rank test) should be performed to confirm that the improvements over the vanilla MoE baseline are not due to chance. Without these additions, the empirical evidence remains anecdotal rather than statistically sound.

Action items.

- **[science]** The paper reports downstream performance improvements (e.g., Table 1, Table 3) and loss reductions (e.g., Figure 1, Table 2) as deterministic facts without reporting statistical significance. Given the stochastic nature of deep learning training, the authors must report standard deviations or confidence intervals across multiple random seeds (e.g., 3-5 runs) to confirm that the observed gains are not due to random variance.

- **[science]** In Section 5.1 (Ablation Studies), the claim that 10 power iterations 'provides no further convergence advantage' is based on a single run comparison (0.002-0.003 loss increase). Without error bars or variance estimates, it is impossible to distinguish a genuine performance drop from noise. The authors should re-run these ablations with multiple seeds to validate the stability of the single-iteration choice.
- **[science]** The sensitivity analysis of hyperparameter C' (Table 4) presents a single validation perplexity value for each setting. To support the claim of 'insensitivity,' the authors should report the variance of these metrics across different random seeds or data shuffles, as small fluctuations in PPL could otherwise be misinterpreted as robustness.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a novel approach to Mixture-of-Experts (MoE) router design with generally clear and professional writing. The logical flow from motivation to methodology and experiments is coherent. However, the text contains several recurring grammatical errors, typos, and awkward phrasings that detract from the overall readability and require correction before publication.

Specifically, there are multiple instances of subject-verb agreement errors. For example, in Section 4.2, the sentence "aggressive alignment disrupt the stability" incorrectly uses the plural verb "disrupt" with the singular subject "alignment"; it should be "disrupts." Similarly, in Section 5, "Extensive experiments validates MPI" should read "validate." In Section 4.2, "it impose no constaint" contains both a verb agreement error ("imposes") and a spelling error ("constraint").

There are also issues with word choice and typos. In Section 4.2, "more challenge tasks" should be "more challenging tasks," and "downstream performance stills improves" should be "still improves." In Section 3.2, the phrase "to to prevent explosion" contains a duplicated word. Additionally, the sentence "while designed to avoid instability, this retraction provides additional benefits" in Section 3.2 suffers from a dangling modifier, as the retraction itself was not the entity that "designed" the system.

Finally, some sentences could be tightened for better clarity. The abstract's statement that "rows of the router matrix compute their similarity" is slightly personified and could be rephrased to clarify that the matrix computes the similarity scores. Addressing these mechanical and stylistic issues will significantly improve the polish of the paper.

Action items.

- **[writing]** In Section 3.2, fix the dangling modifier in 'while designed to avoid instability, this retraction provides...' by rephrasing to 'While designed to avoid instability, the retraction step also provides...'.

- [**writing**] In Section 3.2, remove the duplicate word in 'to to prevent explosion'.
- [**writing**] In Section 4.2, correct the subject-verb agreement error: change 'aggressive alignment disrupt the stability' to 'disrupts'.
- [**writing**] In Section 4.2, change 'more challenge tasks' to 'more challenging tasks'.
- [**writing**] In Section 4.2, fix 'it impose no constaint' to 'it imposes no constraint'.
- [**writing**] In Section 4.2, correct 'downstream performance stills improves' to 'still improves'.
- [**writing**] In Section 5, change 'Extensive experiments validates MPI' to 'validate'.
- [**writing**] In the Abstract, clarify 'rows of the router matrix compute their similarity' to 'the router matrix computes similarity scores between its rows and inputs'.